

Bayesian Methods in Epidemiology

How to update what we believe about disease and risk as new evidence arrives — and why the answer can depend on what we believed beforehand.

Based on the chapter by Leonhard Held · *Handbook of Epidemiology*

ROADMAP

Six Steps, From One Test Result to Mapping Disease Across a Country

- 1 Updating belief with evidence** — a diagnostic test shows how prior belief and new data combine.
- 2 The general recipe** — prior, data, posterior, credible intervals, and where a prior comes from.
- 3 Real epidemiologic data** — estimating a prevalence, a disease rate, and an odds ratio from a case-control study.
- 4 Shortcuts and surprises** — approximating hard problems, and how p-values secretly relate to belief.
- 5 Testing the model itself** — checking assumptions, and weighing competing explanations.
- 6 Pooling information** — smoothing noisy small-area estimates and mapping disease by simulation.

TWO PHILOSOPHIES

Two Honest Ways to Define “Probability”

The Frequentist View

Probability is a long-run frequency — how often an event would occur if an experiment were repeated forever. A disease rate or a drug’s effect is a fixed, unknown number, not something that itself has a probability.

The Bayesian View

Probability is a degree of belief, which can apply to anything uncertain — including a disease rate or a drug’s effect. That belief starts somewhere and is updated as evidence arrives.

Neither view is wrong. They answer different questions — and this chapter is about what the Bayesian view offers epidemiology.

WHY IT MATTERS

Epidemiologists Rarely Start From Nothing

A new study almost never lands in a vacuum. Earlier research, laboratory biology, and expert judgment all point somewhere before the data are even collected. The Bayesian approach gives epidemiologists a formal way to combine that existing knowledge with new evidence, rather than analyzing each study as if nothing were known before it.

Direct answers

It can state, in plain probability terms, how likely a range is to contain the truth — the question people actually want answered.

Strength in small numbers

When data are sparse — a rare exposure, a small geographic area — it lets related groups lend each other statistical strength.

Transparency

Prior assumptions must be stated explicitly, putting them up for scrutiny and debate rather than leaving them hidden.

A FAMILIAR PUZZLE

You Test Positive. How Worried Should You Be?

A test comes back positive for a disease. It is natural to ask: given this result, what is the actual chance I have the disease? The surprising answer is that this depends heavily on something that has nothing to do with the test itself — how common the disease already is. Bayes' theorem is the tool for working this out exactly.

EXAMPLE 1 • A DIAGNOSTIC TEST

A Test That Gets It Right 90% of the Time

If you have the disease...

the test correctly says so **90% of the time**. (This is its *sensitivity*.)

If you don't have the disease...

the test correctly says so **90% of the time**. (This is its *specificity*.)

One more fact matters just as much: only **1% of people** in the population being tested actually have the disease. That number — the disease's *prevalence* — is the “prior” belief we start from before anyone is tested.

EXAMPLE 1 • A DIAGNOSTIC TEST

Why Rare Diseases Trip Up Good Tests

Even a well-made test slips up occasionally. Applied to a population where the disease is rare, that small slip-up rate multiplies against an enormous group of healthy people — producing more false alarms in absolute numbers than the small group of true cases can ever generate correctly.

100

people out of every 10,000 actually have the disease

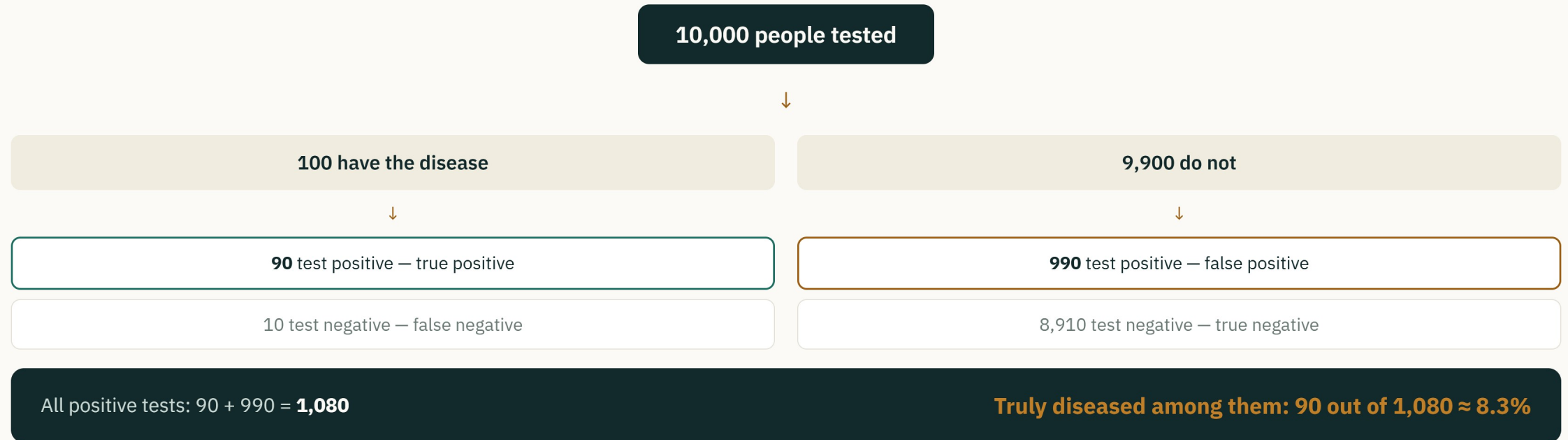
9,900

do not — a group 99 times larger

The next slide traces exactly what happens when the test is run on all 10,000 people.

EXAMPLE 1 • A DIAGNOSTIC TEST

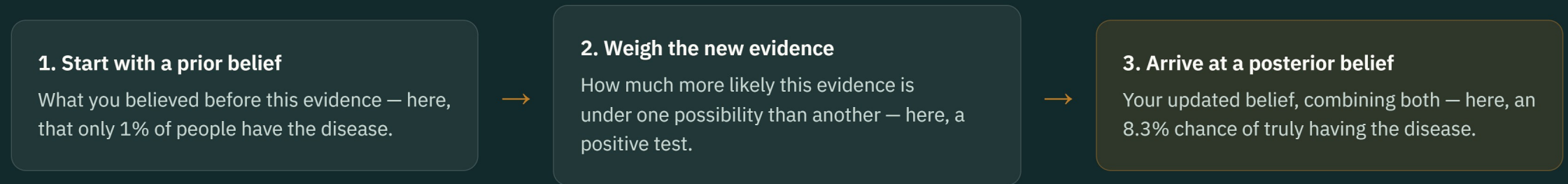
Tracing the Numbers, Step by Step



THE BIG IDEA

Bayes' Theorem, in Plain Language

The tree diagram was not a special trick for this one example — it was Bayes' theorem, worked out by hand. Whenever new evidence arrives, the same three-step logic applies, whatever the question.

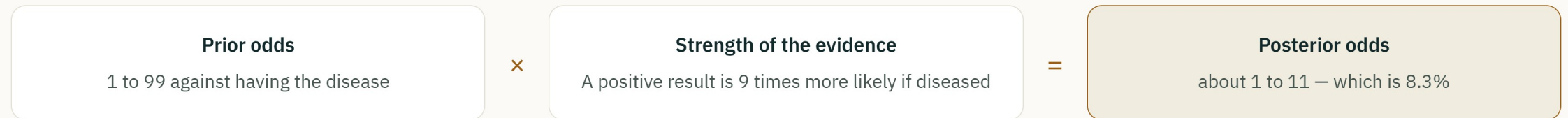


Every Bayesian analysis in this chapter, however complicated, is this same update — run once, or many times as more evidence arrives, with today's posterior simply becoming tomorrow's prior each time, a pattern the chapter calls Bayesian learning.

EXAMPLE 1 • A SHORTCUT

A Shortcut Using Odds

There is a faster route to the same 8.3% answer, using odds instead of percentages. Odds compare two chances directly — a 1-in-100 disease chance is odds of 1 to 99 against.



A negative result works the same way in reverse: it is 9 times more likely if healthy, so it divides the prior odds by 9 instead, landing at about 1 to 891 — roughly a 0.1% chance of disease.

This shortcut also separates two things that are easy to blur together: how strongly the evidence itself points one way — a fact about the test, which never changes — from what you believed beforehand, which is different in every population.

THE GENERAL RECIPE

Three Ingredients: Prior, Data, Posterior

1. Prior

What you believe about the unknown quantity before this study — a full range of possible values, each with its own plausibility, not just one guess.

2. Data (Likelihood)

The new evidence from the current study — how compatible the observed data are with each possible value of that quantity.

3. Posterior

The updated belief after combining the two — again a full range of values, now sharpened by the data.

This works whether the unknown quantity is the chance of a disease, a relative risk, or a rate of new cases per year — anything expressible as a number.

THE GENERAL RECIPE

From Belief to a Range: Credible Intervals

Because the posterior describes a whole range of possible values, it is natural to summarize it with an interval. A 95% credible interval is a range that we can say, directly and literally, has a 95% chance of containing the true value.

How this differs from a traditional confidence interval

A confidence interval's technical meaning is less direct: if the entire study were repeated infinitely many times, 95% of the intervals calculated that way would contain the true value — but any single interval either does or doesn't, and you cannot say it has a 95% chance of being right. A Bayesian credible interval can make exactly that statement.

THE GENERAL RECIPE

Two Ways to Draw the Boundaries

Equal-tailed interval

Leaves exactly the same amount of leftover belief on each side — 2.5% below and 2.5% above, for a 95% interval. Simple to compute, and the natural choice when the posterior is symmetric.

Highest posterior density (HPD) interval

The shortest possible interval that still captures 95% of the belief, so it only ever excludes the least plausible values. Can look noticeably different when the posterior is lopsided.

The two usually agree closely for bell-shaped posteriors, and diverge more only when the distribution is skewed.

CHOOSING A PRIOR

Choosing a Starting Point: Informative vs. Neutral

Every Bayesian analysis needs a prior, and choosing one is a genuine decision, not a technicality.

Informative prior

Built from real outside knowledge — an earlier study, a meta-analysis, or expert opinion. It pulls the answer toward what was already believed, more strongly when the new data are sparse.

Reference (non-informative) prior

Built to have as little influence as possible, letting the new data speak almost entirely for themselves — useful when no credible outside knowledge exists.

Neither is automatically right. An analysis can, and often should, be run under more than one prior, to see how much the choice actually matters.

CHOOSING A PRIOR

The Search for a Truly Neutral Prior

Being neutral is harder than it sounds: a prior that looks flat and uninformative in one way of expressing a number can look strongly opinionated in another. A mathematician named Harold Jeffreys developed a general method for building a neutral, or reference, prior — now called *Jeffreys' prior* — using a rule independent of the scale the quantity is expressed on.

A curious side effect

Some of these deliberately neutral priors are so spread out that they no longer add up to a genuine probability distribution at all — they are called *improper* priors. That is usually fine for estimating a single quantity, since real data can still pull the result into a proper, usable posterior. But it generally cannot be used for comparing rival models, since that comparison needs a genuine starting probability distribution to work at all.

ESTIMATING A PREVALENCE

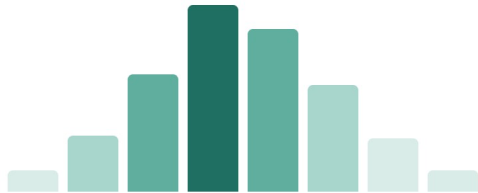
Estimating How Common a Disease Is

Earlier, we treated the disease's 1% prevalence as a known fact. In reality, prevalence is itself usually estimated from a sample — testing some number of people and finding a proportion positive — and that estimate carries its own uncertainty. This is the simplest possible case for applying the full Bayesian recipe: prior, data, posterior.

ESTIMATING A PREVALENCE

A Natural Mathematical Pairing

When the unknown quantity is a proportion — a value that must fall between 0% and 100% — a flexible curve called a *beta distribution* is the natural choice for expressing belief about it. Combining a beta belief with real yes/no data always produces another beta curve back, with no complex computation required. This convenient pairing is called *conjugacy*.



Starting from a flat, equally open-minded belief — the same starting point Thomas Bayes himself used in his original 1763 essay — and then observing 3 successes in 8 new trials, belief updates to a curve peaking around 0.4, with a 95% credible interval running from about 0.14 to 0.70.

EXAMPLE 1 • REVISITED

Back to the Diagnostic Test: What If We're Not Sure About Prevalence?

So far, the disease's 1% prevalence was treated as a known fact. In reality it is normally itself estimated from a small survey. Suppose a prevalence survey tested 100 people and found just 1 with the disease.

1 in 100

people tested positive in the prevalence survey

about 2%

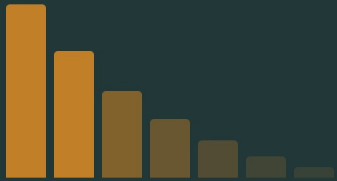
updated best-guess prevalence — but with real uncertainty around it, not a fixed number

Once prevalence itself is a range of possibilities rather than one fixed number, what happens to the 8.3% chance of disease after a positive test?

EXAMPLE 1 · REVISITED

Simulating Thousands of Possible Truths

There is no simple direct formula once an uncertain prevalence feeds into the answer. So instead, the analysis *simulates*: draw one plausible prevalence value from its updated belief, calculate the resulting chance of disease given a positive test (holding the test's 90%/90% accuracy fixed), and repeat this ten thousand times.



The honest answer is a whole spread of plausible values, not one number: a posterior mean chance of disease of **14.5%**, most likely value 9.6%, and a 95% credible interval running from about 2% to 34%.

That is much higher, and far more honestly uncertain, than the tidy 8.3% obtained earlier by treating prevalence as exactly known.

EXAMPLE 2 • AN ELEVATED CANCER RISK?

Does an X-ray Procedure Raise Cancer Risk?

In the mid-20th century, women being treated for tuberculosis were often monitored with a repeated chest X-ray procedure called fluoroscopy. Researchers later asked whether that repeated radiation exposure measurably raised these women's risk of developing breast cancer.

41

breast cancer cases actually observed among the x-rayed women

23.3

cases that would have been expected at typical rates for their age

The question: how much higher is the true underlying risk, and how confident can we be in that number?

ESTIMATING A RATE

A Natural Pairing for Rates

Just as a beta curve suits a proportion, a different flexible curve — a *gamma distribution* — is the natural match for a rate, such as how many times more or less common a disease is than expected. Pairing it with count data again produces a clean, directly computable posterior.

The same weighted-average logic as before

The resulting estimate always blends the starting belief with the observed data, weighted by how much each one represents. A belief built from just a handful of imagined cases barely resists real data drawn from dozens of actually observed ones.

EXAMPLE 2 • COMPARING PRIORS

Three Experts, Three Priors

To see how much the choice of prior actually matters, the analysis was run under three different starting beliefs about the true relative risk, before folding in the same data — 41 cases observed against 23.3 expected.

Starting belief (prior)	Resulting estimate	95% credible interval
Moderate: risk probably between half and double the norm	1.55	1.15 – 2.01
The same belief, rebalanced to be equally skeptical either way	1.61	1.19 – 2.08
A mathematically neutral reference prior	1.78	1.28 – 2.36

All three posterior analyses suggest an elevated rate ratio, most likely 50–80% above baseline, though with substantial uncertainty — and the more mathematically neutral prior nudges the estimate a little higher.

EXAMPLE 2 • COMPARING PRIORS

What Changed, and What Didn't

Every estimate above is really a weighted average of two things: what was believed beforehand, and what the 41-versus-23.3 data showed on their own. The starting belief behaved as if built from the equivalent of only about a quarter as much evidence as the real data provided — so the real data dominated the answer, but not completely.

A general, reassuring property

As real evidence accumulates, the influence of the starting belief fades, and analysts who began with different opinions are pulled toward the same conclusion. With very little data, by contrast, the choice of prior can matter a great deal — which is exactly why it deserves to be stated openly rather than left implicit.

EXAMPLE 3 • A CASE-CONTROL STUDY

Power Lines and Childhood Leukemia

A long-debated public health question: does living near high-voltage power lines, and the resulting exposure to magnetic fields, raise a child's risk of leukemia? Because childhood leukemia is rare, a case-control study is the practical design — find children who already have leukemia, find similar children who don't, and compare how each group's past exposure compares.

EXAMPLE 3 • THE DATA

Reading the Data

Exposure was defined as a measured magnetic field above 3 milligauss. Among the children with leukemia, a larger share had been exposed than among the healthy comparison children:

	Exposed	Unexposed
Cases — children with leukemia	3	5
Controls — children without	33	193

Using flat, equally open-minded starting beliefs for each group’s true exposure rate, and simulating ten thousand plausible combinations, the analysis arrives at a posterior belief about the odds ratio between the two groups.

EXAMPLE 3 · A CLEVER RESULT

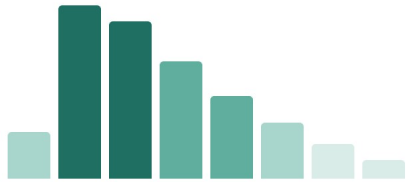
Cornfield's Clever Shortcut

A case-control study can only directly compare exposure rates between cases and controls — the *exposure* odds ratio. But the question everyone actually wants answered runs the other way: how much more likely is disease among the exposed than the unexposed — the *disease* odds ratio.

In 1951, the statistician **Jerome Cornfield** proved, using Bayes' theorem, that these two odds ratios are mathematically identical. This is why a case-control study can recover the disease odds ratio even though sampling begins with cases and controls.

EXAMPLE 3 • THE RESULT

The Resulting Estimate



Simulating ten thousand plausible combinations of the two groups' exposure rates gives a posterior mean odds ratio of **4.7**, with a 95% credible interval running from **0.9 to 14.3**.

That is a wide range, and it includes values close to 1 — no association at all. So despite the striking-looking average of 4.7, the honest conclusion at this stage is real uncertainty, not proof of an effect.

WHEN THE MATH GETS HARD

Approximating the Posterior

Not every problem has a clean, directly computable answer like the ones so far. When a model grows more complex — adjusting for other factors alongside the main question, say — an exact posterior can become mathematically intractable. A practical shortcut helps: treat the single best estimate a standard analysis already produces as if it were itself one clean, bell-curve-shaped piece of evidence, and update from there.

WHEN THE MATH GETS HARD

Treating the Posterior Like a Bell Curve

Most evidence curves are approximately bell-shaped once enough data accumulate — especially on a scale, like a logarithm of relative risk, that is not artificially restricted to positive numbers only. A bell-shaped starting belief, combined with bell-shaped evidence, again produces a clean, directly computable, bell-shaped posterior — no different in spirit from the pairings already seen.

Seeing it in the breast-cancer data

The best single estimate of the (logged) relative risk in Example 2 was 0.57, with an approximate uncertainty of 0.16 built from the 41 cases observed. The bell-curve approximation sits almost exactly on top of the true evidence curve near this peak, drifting away only well outside it.

EXAMPLE 2 • THE SHORTCUT APPLIED

Applying the Shortcut

Applying it to the breast-cancer example gives a posterior mean relative risk of **1.62**, 95% credible interval **1.21 to 2.12** — nearly identical to the 1.55 [1.15, 2.01] found earlier with the exact, non-approximated calculation. The shortcut and the exact math agree closely.

A remarkable special case

Push the starting belief to be completely flat — infinitely uncertain — and the Bayesian answer becomes numerically identical to the traditional confidence interval a frequentist analysis would report. Seen this way, a standard frequentist analysis is a Bayesian analysis with one particular, rather unrealistic, starting belief: one that gives equal weight even to extreme, implausible values of relative risk.

RETHINKING THE P-VALUE

Sometimes a P-Value Matches a Bayesian Tail Probability

Statisticians have long used a “p-value” — asking how unusual results this extreme would be if the null hypothesis were true. In one special case, under a particular neutral starting belief, that traditional p-value can turn out strikingly close to a genuine Bayesian posterior probability — the direct, literal probability that the effect lies on one side of “no effect at all.” Change the starting belief, and that closeness disappears.

Traditional one-sided p-value (breast-cancer data)

0.00057

A similar Bayesian posterior probability

0.00042

Two entirely different schools of thought, starting from entirely different premises, landing on startlingly similar numbers.

A HISTORICAL DISCOVERY

A 19th-Century Rediscovery

This connection is not new. In **1877**, the German physician Carl von Liebermeister proposed exactly this kind of Bayesian approach for comparing two groups in a medical study — over fifty years before the modern p-value was formalized. Working from flat, open-minded starting beliefs, he showed how to calculate the posterior probability that one group's true rate exceeded another's.

An unexpected twin

His calculation turns out to be mathematically equivalent to a well-known modern test — Fisher's exact test — applied to a slightly adjusted version of the same table. Later researchers found that Liebermeister's version actually performs a little better, on average, than Fisher's own test.

EXAMPLE 3 • CONTINUED

Back to the Leukemia Study

Applying this logic to the power-line data gives several honest answers to “how likely is it that there is truly no effect,” depending on what counts as a fair starting belief:

0.108 Fisher’s traditional one-sided p-value

0.036–
0.048 Lieberman’s Bayesian posterior probability, from a flat starting belief

0.127 Using a more realistic starting belief, allowing the odds ratio to plausibly range from a quarter to four times

Which number is “right” depends entirely on what starting belief is defensible. The p-value itself uses no prior at all; the Bayesian interpretation does — and, as these three numbers show, different priors can give different posterior probabilities from the very same data.

A SECOND USE FOR THE POSTERIOR

Prediction: What Might We Observe Next?

Bayesian inference is not only for pinning down an unknown quantity. Once the posterior captures what remains uncertain, it can be averaged over to forecast what a new patient, a new trial, or the next observation might actually look like.

Posterior predictive

What comes next, now that this data has been seen. Averages over the updated, sharpened uncertainty in the parameter.

Prior predictive

What would have been expected before this data existed at all. Averages over the original starting belief instead.

That second idea, the prior predictive, turns out to matter enormously: it is the key ingredient in checking whether a starting belief fits the data at all, and in comparing rival models against each other.

CHECKING THE MODEL'S ASSUMPTIONS

Can the Model Predict Its Own Data?

Before trusting any conclusion, it is worth a basic sanity check: does the starting belief, combined with the kind of data expected, make the data actually observed look at least plausible? If the real result would have been a shock under the assumed starting belief, something is wrong — either the prior was a poor choice, or the data are telling us something the model is not capturing.

A formal “surprise score”

This works by asking what data the model would have predicted before seeing anything, then checking where the actual result falls within that predicted spread. The statistician George Box formalized this check in 1980: a small score flags a real conflict between the prior and the data.

EXAMPLE 3 · CHECKING THE MODEL

Applying This to the Leukemia Example

Running this surprise check on the leukemia study's informative starting belief — that the true odds ratio was probably between one-quarter and four — against the data actually observed gives a score of **0.22**: comfortably unremarkable. The starting belief and the real-world data were reasonably compatible; nothing here suggests the assumed prior was a poor fit to reality.

COMPARING HYPOTHESES, NOT JUST ESTIMATING

Is There Really an Effect at All?

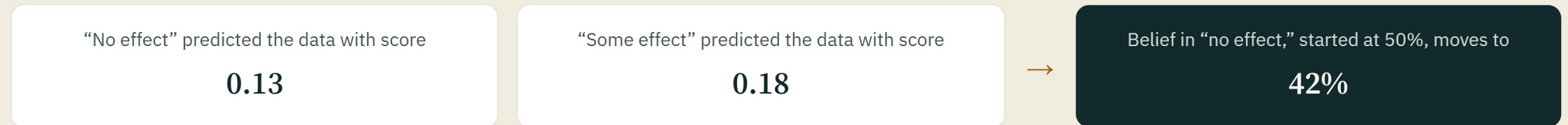
Every analysis so far has assumed some effect exists, and asked how big it is. A different, equally important question is whether there is good reason to believe any effect exists at all, compared to the simplest explanation: no effect whatsoever.

Bayesian model comparison treats both possibilities — an effect, or no effect — symmetrically, weighing the evidence for each. Traditional hypothesis testing cannot do this: it can only ever reject the “no effect” explanation, or fail to reject it. It has no mechanism for actively confirming that “no effect” is the better explanation.

COMPARING HYPOTHESES, NOT JUST ESTIMATING

The Bayes Factor: Weighing Two Hypotheses

A Bayes factor compares how well two competing explanations predicted the data that were actually observed. Multiplied by how strongly each was believed beforehand, it gives an updated verdict: the posterior odds between the two explanations.



Applied to the leukemia data, comparing "no effect" against Greenland's informative belief in some effect: the resulting Bayes factor barely moves the needle.

COMPARING HYPOTHESES, NOT JUST ESTIMATING

A Puzzling Disagreement

A standard two-sided significance test on this same leukemia data gives a p-value of about 0.10 — often read informally as “borderline, but suggestive of a real effect.”

Traditional p-value

≈ 0.10

often read as suggestive

Posterior probability of “no effect”

42%

barely below the 50% starting point

This is a widely documented, important disconnect: a p-value in the “suggestive” range can correspond to only the weakest possible shift in genuine belief.

COMPARING HYPOTHESES, NOT JUST ESTIMATING

A Built-in Safeguard

How far can this disconnect go? Statisticians have derived a “minimum Bayes factor” — the strongest possible evidence against “no effect” that any prior in the chosen class could produce, given a particular p-value. It is a built-in ceiling on how much evidence a p-value can ever represent.

Applied to the leukemia data

Even in the most extreme, unrealistic best case for “some effect” — a prior concentrated exactly on the observed estimate — the posterior probability of “no effect” cannot fall below about **20%**. The more realistic Sellke–Bayarri–Berger bound, still generous to “some effect,” puts the true floor closer to **38%**.

The general lesson, true well beyond this one example: a small p-value, on its own, almost never means what casual intuition suggests.

EXAMPLE 4 • DISEASE MAPPING

Mapping Lip Cancer Across Scotland

A different kind of challenge: estimating disease rates not for one population, but separately for each of **56 regions** across Scotland, based on the lip cancer cases recorded in each. Some regions have plenty of data; many small, rural ones have only a handful of cases, making their raw rates wildly unreliable on their own.

BORROWING STRENGTH ACROSS GROUPS

The Problem With Small Areas

A region with only one or two recorded cases can show a raw rate that looks dramatically high or low purely by chance — not because the true underlying risk is actually unusual there. Treating each region's raw count in isolation produces a noisy, unreliable map, dominated by statistical accidents in the smallest regions.

BORROWING STRENGTH ACROSS GROUPS

Shrinkage: Letting Areas Learn From Each Other

The fix is to let all 56 regions share information. Rather than giving each region its own unrelated starting belief, this approach estimates *one shared* starting belief directly from the pattern across all regions together, then blends each region's own raw data with that shared belief.

In Scotland's lip cancer data, this shared belief settled on an average relative risk of about **1.42**. Every region's estimate was then pulled — “shrunk” — from its own noisy raw rate toward this common value: more so for small, data-poor regions, less so for large, data-rich ones.

SIMULATING THE ANSWER

When There's No Shortcut Left: Simulating an Answer

Shrinking every region toward one shared national average is a good start, but it treats a city and a remote island the same way, ignoring geography entirely. A more refined model lets each region learn directly from its immediate neighbors — a genuinely high-dimensional problem too complex for any clean formula.

Simulating a way through

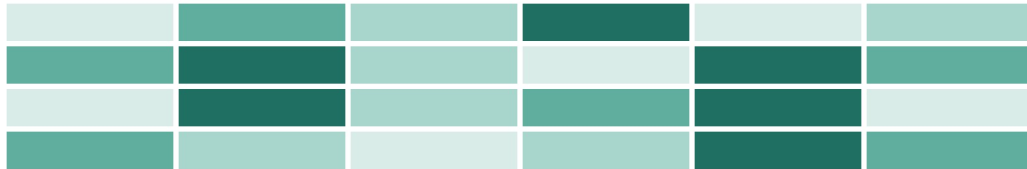
The solution is to build a long, self-correcting chain of guesses that, after enough steps, settles into a pattern matching the true posterior belief — a technique called *Markov chain Monte Carlo*. One common version, a *Gibbs sampler*, updates one region's estimate at a time, each time using the current guesses for all the others. For Scotland's map, this chain ran for 100,000 steps, discarding the first 10,000 while it settled in.

EXAMPLE 4 • CONCLUDED

Adding Geography: Neighbors Inform Neighbors

Rather than pulling every region toward one national number, this refined model pulls each region toward its immediate geographic neighbors' values instead. The result is a noticeably smoother map, where risk changes gradually across the country instead of jumping erratically from one county to the next.

Shrunk toward one national average



Smoothed toward geographic neighbors



Two Philosophies, One Goal

Both schools of thought are trying to do the same thing: draw honest conclusions from imperfect data. They just define the basic question differently.

Frequentist statistics asks

If this study were repeated many times, how often would a method like this one be right?

Bayesian statistics asks

Given what actually happened, how likely is each possible truth?

The Author's View: Complementary, Not Competing

Leonhard Held, the chapter's author, closes with a personal reflection: a frequentist analysis based on likelihood alone can be seen as a limited form of Bayesian inference, used in the absence of prior knowledge, and it typically leads to numerically similar estimates. But increasingly, stating a prior explicitly is seen as valuable, since it avoids the unrealistic assumptions hidden inside a purely frequentist analysis. Empirical Bayes methods, which estimate the prior from the data itself, sit as a compromise between the two traditions.

Where the two can diverge sharply is hypothesis testing — as this chapter's own leukemia example showed, the evidence against “no effect,” measured by a Bayes factor, is often far weaker than a p-value alone would suggest.

SOURCE & ATTRIBUTION

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